RECOVER VMs WITH GOOGLE BACKUP AND DR SERVICE

PROJECT DESCRIPTION

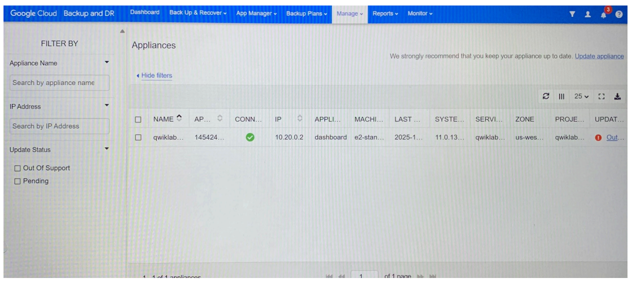
In this lab activity I discovered and protected a compute Engine Instance and mounted a fully functional new compute Engine Instance from the backup image to a new location.

SCENARIO

I am to assist with implementing recovery actions to restore affected virtual machines in a security incident.
Here's how I did this;

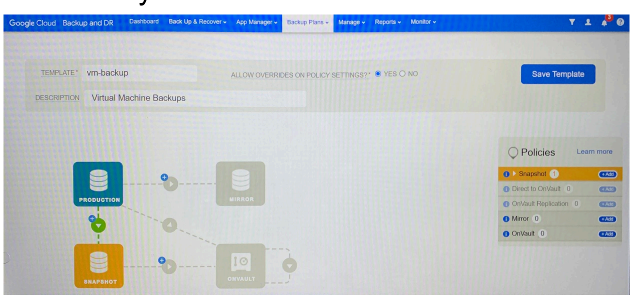
First Step: Connect to the Backup and DR Console

Here, I successfully installed the management server and the Backup and Recovery server.



Second Step: Create a Backup Plan template

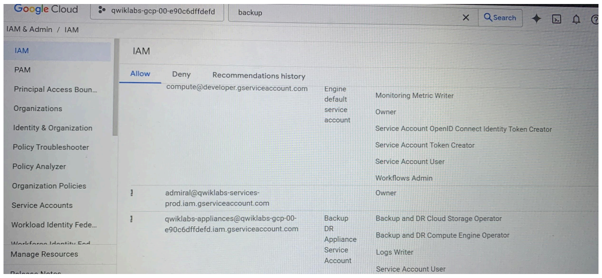
Backup plan templates are composed of backup policies. In policies, I defined when to run a backup, how frequently to run a backup, how long to retain the backup image for days, weeks, months or years.



I kept the Backup and DR management console open in a new tab for the entire lab.

Third Step: Validate the Backup and Recovery Appliance Service Account Permissions

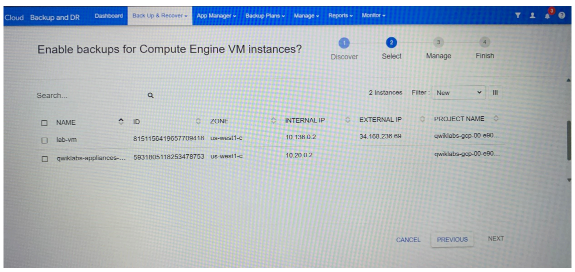
Here, I viewed the required IAM roles of the backup/recovery appliance to verify that it has the correct IAM roles.



In the **role** column, I noticed that the **Backup and DR Cloud Storage Operator role** is already assigned.

Fourth Step: Discover and Add Compute Engine Instances to the Management Console

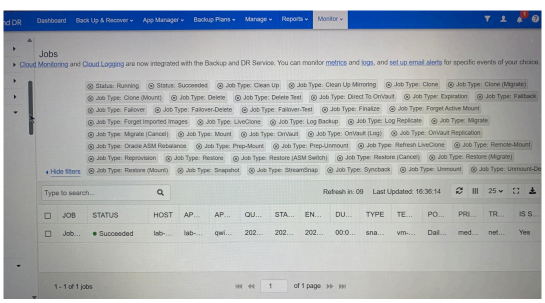
Here, I used the **onboarding wizard** to onboard my Compute Engine Instances. Onboarding an Instance means I attach the template to the Instance.



A summary of changes appear and provides the following information;

1. Instance Name : **lab–vm**
2. Appliance : **Qwiklabs–appliance**
3. Action: Apply a backup template

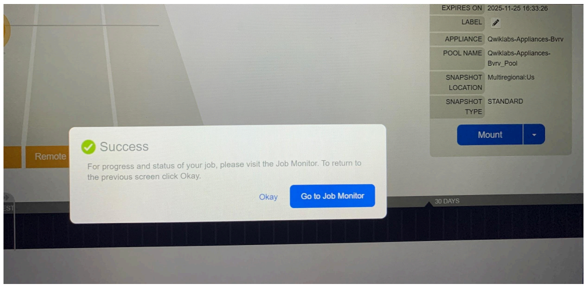
I selected **lab–vm** compute engine instance for backup.
After onboarding is complete, the status is a green check. This means the policy template is attached to the selected VM.

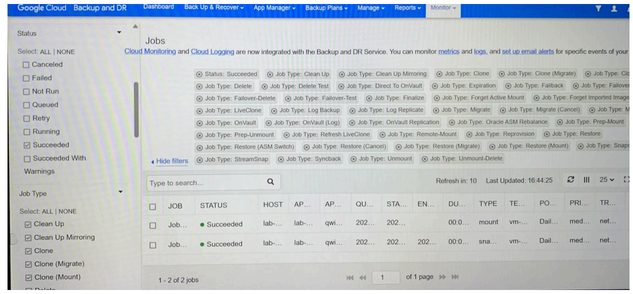


I can monitor the progress of the backup job. When the job is finished, I have an image that I can restore if needed.

Fifth Step: Restore a Compute Engine Instance

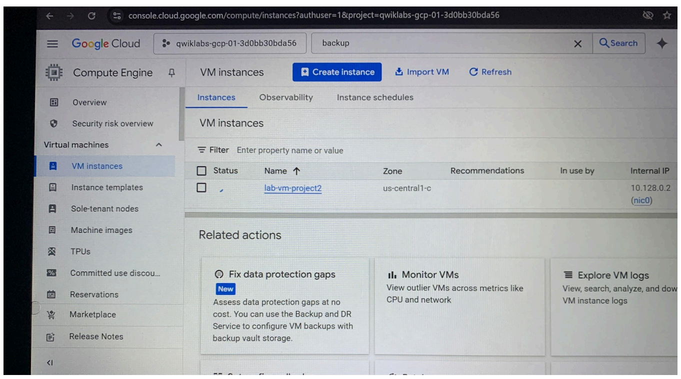
Now that I have an image of my Compute Engine Instance. In this step, I created a brand new Compute Engine Instance using the backup image that I created in the previous step.
When both jobs have a succeeded status, I have the Computer Engine Instance.





Final Step: Restore a compute Engine Instance to an alternate Project.

Here in this final step, I restored a Computer Engine Instance using the backup template I created but this time to a different project.
I can also create a brand new Compute Engine Instance in a different project from backup images.



CONCLUSION

I successfully used Google backup and DR service to create a backup template and then applied it to two compute Engine instances.
On completion of this lab, I have shown how to prepare for issues with VMs and the service.
When a device malfunctions, I can use Backup and DR service to restore malfunctioning devices across multiple Google Cloud projects.